(1) Show that the equation of the line passing through $A(x_1, y_1)$ and parallel to the line $ax+by+c=0$ is $a(x-x_1)+b(y-y_1)=0$. Hence find the equation of the line passing through $(1,1)$ and parallel to the line $15x+8y+1947=0$.

Solution : Slope of the line $ax+by+c=0$ is $-\frac{a}{b}$.

Required line is parallel to this line.

$\therefore$ The slope of the required line is also $-\frac{a}{b}$. And it passes through $A(x_1, y_1)$.

The equation of the required line is $y-y_1=-\frac{a}{b}(x-x_1)$

$\therefore\ by-by_1=-ax+ax_1$

$\therefore\ a(x-x_1)+b(y-y_1)=0$

Now the equation of the line parallel to $15x+8y+1947=0$ and passing through (1, 1) is

$15(x-1)+8(y-1)=0$

$\therefore\ 15x+8y-23=0$

(2) Show that the equation of the line having slope m and making x-intercept d is given by $y=m(x-d)$. Find the y-intercept of this line.

Solution : The x-intercept of the line is d. Therefore it cuts the X axis in $(d, 0)$.

Slope of the line is m.

$\therefore$ It's equation of the line is $(y-y_1)=m(x-x_1)$

$\therefore$ $(y-0)=m(x-d)$

$\therefore$ $y=m(x-d)$ ie. $y=mx-md$

To find it's y-intercept,

The equation is

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$y = mx - md$

This equation is of the From $y = mx + c$, where c is the y-intercept of the line.

$\therefore c = -md$

$\therefore$ y-intercept $= -md$

(3) A line makes intercepts h and k on the co-ordinate axes. If p is the length of the perpendicular drawn from the origin to the line then show that $\frac{1}{h^2}+\frac{1}{k^2}=\frac{1}{p^2}$.

Solution : Line makes intercepts h and k on the co-ordinate axes. Its equation is $\frac{x}{h}+\frac{y}{k}=1$.

$kx+hy-hk=0$

The length of the perpendicular drawn from the origin to the line $kx+hy-hk=0$ is

$$p=\left|\frac{-hk}{\sqrt{k^2+h^2}}\right|$$

$$\therefore\ p^2\left(k^2+h^2\right)=h^2k^2$$

$$\therefore\ \frac{h^2+k^2}{h^2k^2}=\frac{1}{p^2} \qquad \therefore\ \frac{1}{h^2}+\frac{1}{k^2}=\frac{1}{p^2}.$$

(4) Show that there are two lines which pass through the point $A(3,7)$ and the sum of whose intercepts on the co-ordinate axes is zero. Draw the rough sketch of these two lines. -

Solution : Let slope of one of the lines be m.

Its equation is $y-7=m(x-3)$

$mx-y+(7-3m)=0 \Rightarrow mx-y=3m-7 \Rightarrow \frac{x}{\frac{3m-7}{m}}+\frac{y}{7-3m}=1$

Intercepts made by this line on the co-ordinate axes are $\frac{3m-7}{m}$.

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and $1-3m$

Their sum is $\frac{3m-7}{m}+7-3m$

But it is given that the sum of intercepts is zero.

$\therefore \frac{3m-7}{m}+7-3m=0$

$\therefore 3m-7+7m-3m^2=0$

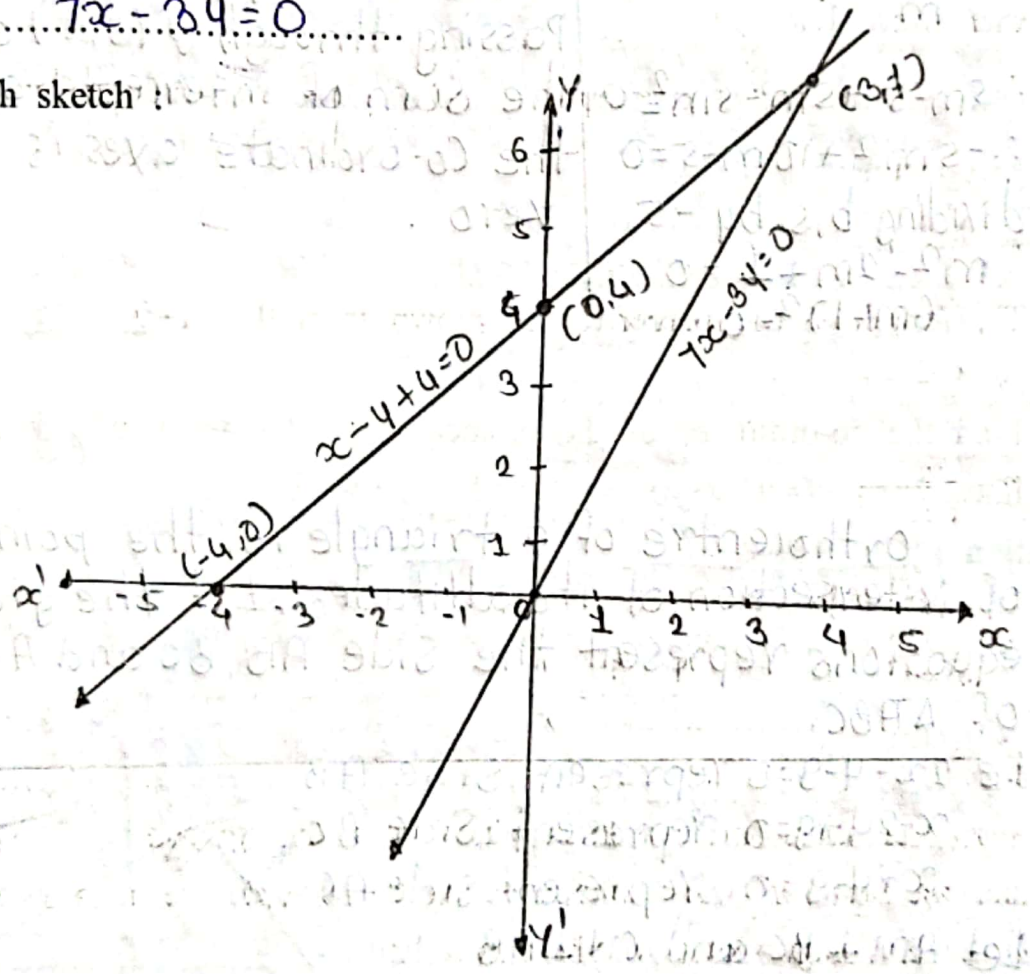
$\therefore 3m^2-10m+7=0 \Rightarrow 3m^2-3m-7m+7=0$

$\therefore m = 1$ Or $m = 7/3$

As m has two values, there are two lines.

Their equations are $x-y+4=0$ and $7x-3y=0$

Rough sketch :



(5) Find the number of lines which pass through the point $B(5,5)$ and the sum of whose intercepts on the co-ordinate axes is zero.

Solution : Let slope of one of the lines be m

Its equation is $y-5=m(x-5)$

$y-5=mx-5m$

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$\therefore 5x - y = 5m - 5$

$\therefore \frac{5x}{5m-5} - \frac{y}{5m-5} = 1$

$\therefore \frac{x}{\frac{5m-5}{m}} + \frac{y}{5-5m} = 1$

$\therefore$ x-intercept $= \frac{5m-5}{m}$

y-intercept $= 5-5m$

Sum of intercepts $= 0$

$\frac{5m-5}{m} + 5-5m = 0$

$\therefore 5m-5+5m-5m^2 = 0$

$\therefore -5m^2+10m-5=0$

dividing b.s. by -5

$m^2-2m+1=0$

$(m-1)^2=0$

$\therefore m-1=0$

$\therefore m=1$

The equation of line having slope $m=1$ and Passing through $(5,5)$

$y-5=1(x-5)$

$y-5=x-5$

$\boxed{x-y=0}$

Since $m=1$ is the only value There is only one line Passing through $(5,5)$ and the sum of intercepts on the Co-ordinate axes is zero

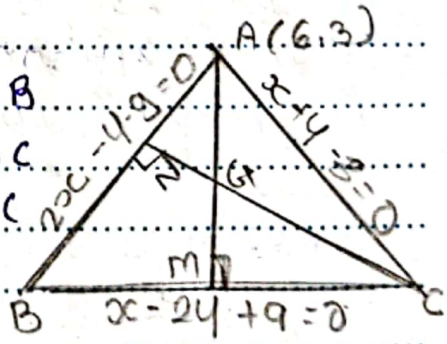
(6) Find the co-ordinates of the orthocenter of the triangle formed by lines $2x-y-9=0$, $x-2y+9=0$ and $x+y-9=0$.

Solution : Orthocentre of a triangle is the point of intersection of its altitudes. Let the given equations represent the side AB, BC and AC of ΔABC.

i.e $2x-y-9=0$ represent side AB

$x-2y+9=0$ represent side BC

$x+y-9=0$ represent side AC



Let $AM \perp BC$ and $CN \perp AB$

Let AM and CN intersect at G

Then G is the orthocentre of ΔABC

Now solving equation of sides AB and AC simultaneously we get co-ordinate of A

$A \equiv (6,3)$

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Solving equations of AC and BC,
we get co-ordinates of C
$\therefore C \equiv (3, 6)$
Now slope of line BC i.e. $x - 2y + 9 = 0$
is $m = \frac{1}{2}$
Slope of AM $\perp$ line BC is $-\frac{1}{m} = -2$

Equation of line AM having slope $m = -2$ and
Passing through the point $A(6, 3) \equiv (x_1, y_1)$ is

$$y - y_1 = m(x - x_1)$$
$$y - 3 = -2(x - 6)$$
$$y - 3 = -2x + 12$$
$$\boxed{2x + y = 15} \quad \text{——} (1)$$

Similarly slope of line AB $2x - y - 3 = 0$ is 2.
Slope of CN $\perp$ line AB is $-1/2$
$\therefore$ Equation of line CN is

$$y - 6 = \frac{-1}{2}(x - 3)$$

$$2y - 12 = -x + 3$$
$$\boxed{x + 2y = 15} \quad \text{——} (2)$$

Solving equn (1) and (2) Simultaneusly
equn (1) − 2 × equn (2)

$$2x + y = 15$$
$$-\ 2x \pm 4y = 30$$
$$-3y = -15$$
$$\therefore \boxed{y = 5}$$

Put $y = 5$ in equn (2)

$$x + 10 = 15$$
$$\therefore \boxed{x = 5}$$

$\therefore$ Orthocentre is $(5, 5)$

Sign of Teacher :

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